

MANGALIA, Romania

WMO: 154990

Lat: 43.8161N

Lon: 28.5875E

Elev: 7

StdP: 101.24

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.8	-5.7	-11.7	1.4	-7.7	-9.3	1.7	-5.3	16.1	3.3	14.0	4.6	4.6	290	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	28.6	23.4	27.7	23.3	27.0	23.0	25.6	27.1	24.8	26.6	24.0	26.0	3.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.3	26.7	24.2	19.1	26.1	23.4	18.2	25.5	78.8	27.0	75.4	26.8	72.4	26.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.6	7.9	-11.8	31.7	2.2	1.7	-13.4	32.9	-14.7	33.9	-15.9	34.9	-17.5	36.1		
(5)			-12.1	27.2	2.1	1.2	-13.6	28.1	-14.9	28.8	-16.1	29.4	-17.6	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	2.1	4.1	6.7	10.7	15.8	20.9	22.6	22.8	19.3	13.5	8.5	3.8
	DBStd	8.10	4.30	4.09	3.50	3.20	2.77	2.42	1.76	2.22	3.16	3.45	4.78	4.31
	HDD10.0	849	246	169	115	29	1	0	0	0	0	11	83	196
	HDD18.3	2514	503	399	361	231	87	6	1	1	27	153	295	451
	CDD10.0	1799	1	3	12	49	180	327	389	398	279	119	38	3
	CDD18.3	423	0	0	0	0	8	83	132	141	57	3	0	0
	CDH23.3	2347	0	0	2	1	17	356	780	1006	181	4	0	0
	CDH26.7	252	0	0	0	0	2	39	63	136	13	0	0	0
Wind	WSAvg	3.3	3.2	3.5	3.6	3.2	3.1	3.0	3.0	3.3	3.6	3.7	3.4	3.3
Precipitation	PrecAvg	463	37	22	33	34	45	45	38	30	53	49	35	39
	PrecMax	803	114	59	91	113	106	172	133	96	157	164	85	125
	PrecMin	250	3	2	1	3	8	3	1	2	5	2	3	4
	PrecStd	124	29	15	21	26	29	37	31	28	41	35	21	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	16.9	20.6	22.4	25.3	29.3	29.2	30.7	28.6	23.8	20.6	15.0
		MCWB	9.5	12.1	11.1	14.5	18.4	21.6	22.6	24.8	18.3	19.4	15.7	11.9
	2%	DB	10.1	13.4	16.3	18.7	22.8	27.6	28.1	28.9	26.0	21.2	17.8	13.2
		MCWB	8.3	10.3	10.4	13.5	18.1	23.0	23.0	24.7	21.5	17.9	15.0	11.3
	5%	DB	8.5	10.7	13.4	16.7	21.4	26.6	27.3	28.0	25.1	19.6	16.3	11.4
		MCWB	7.4	8.3	9.2	13.1	17.9	22.6	23.0	23.8	21.7	17.1	14.8	9.9
	10%	DB	7.4	9.0	11.0	15.0	20.2	25.4	26.6	27.3	24.2	18.4	15.0	9.8
		MCWB	6.6	7.2	8.3	12.6	17.3	21.8	22.9	23.4	21.2	16.2	13.8	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	12.9	11.7	16.0	20.9	25.7	26.1	27.6	24.7	21.1	17.1	13.1
		MCDB	12.5	15.9	16.3	18.8	22.6	27.0	27.1	29.2	25.4	22.6	18.1	14.3
	2%	WB	8.7	10.7	10.9	14.8	19.8	24.6	25.2	25.9	23.8	18.9	16.2	11.7
		MCDB	9.5	13.4	15.1	17.4	21.5	26.3	26.6	27.6	24.7	20.0	17.0	12.6
	5%	WB	7.7	8.8	9.8	13.8	18.7	23.6	24.4	25.1	22.9	17.9	15.1	10.1
		MCDB	8.3	10.2	12.4	16.1	20.5	25.4	26.2	27.2	23.9	19.0	16.3	11.2
	10%	WB	6.8	7.6	8.9	12.7	17.7	22.6	23.7	24.2	22.1	16.9	13.7	8.5
		MCDB	7.3	8.5	10.4	14.5	19.6	24.4	25.8	26.6	23.6	18.0	14.9	9.6
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.7	6.3	6.1	6.2	6.8	7.1	7.1	7.1	6.4	5.7	6.1
		MCDBR	6.8	8.9	10.7	9.0	8.9	8.5	8.1	7.7	7.8	7.6	7.0	7.6
		MCWBR	5.1	6.0	5.8	4.9	5.5	5.3	5.7	6.3	5.8	5.3	4.8	5.7
	5% WB	MCDBR	5.7	8.3	8.3	7.5	7.3	7.0	6.9	6.8	6.4	6.6	5.9	7.1
MCWBR		4.8	5.8	5.2	5.0	5.5	5.7	5.7	6.5	5.6	5.5	4.6	5.5	
Clear-Sky Solar Irradiance	taub	0.357	0.386	0.417	0.456	0.429	0.431	0.441	0.451	0.430	0.399	0.376	0.358	
	taud	2.319	2.239	2.183	2.094	2.258	2.291	2.273	2.237	2.265	2.318	2.321	2.322	
	Ebn at Noon	746	787	812	810	846	843	829	806	791	765	716	703	
	Edh at Noon	84	108	129	153	133	129	130	131	118	99	84	77	
All-Sky Solar Radiation	RadAvg	1.30	2.07	3.36	4.86	6.17	6.74	6.71	6.00	4.30	2.68	1.55	1.16	
	RadStd	0.20	0.26	0.35	0.47	0.35	0.40	0.36	0.30	0.34	0.25	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-123	+106	+65

Nomenclature: See separate page